

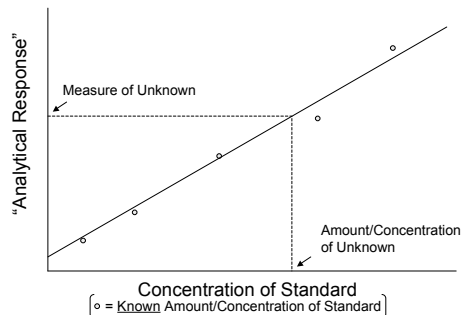
Lecture #5 - Overview

Statistics - Part 3

Statistical Tools in Quantitative Analysis

- The Method of Least Squares
- Calibration Curves
- Using a Spreadsheet for Least Squares

Calibration Curves



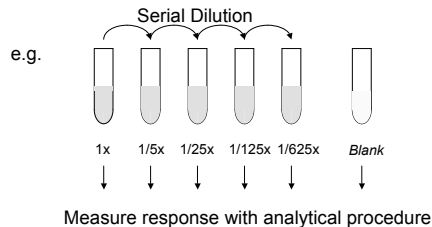
Construction of Calibration Curves

Standard Solutions = "Solutions containing known concentrations of analyte(s)"

Blank Solutions = "Solutions containing all the reagents and solvents used in the analysis, but no deliberately added analyte"

Construction of Calibration Curves

Step 1: Prepare known samples of analyte covering a range of concentrations expected for unknowns. Measure the response of the analytical procedure for these standards.



Construction of Calibration Curves

Step 1: Prepare known samples of analyte covering a range of concentrations expected for unknowns. Measure the response of the analytical procedure for these standards.

Step 2: Subtract the (average) response of the **blank** samples from each measured standard to obtain the corrected value.

$$\text{Corrected} = \text{Measured} - \text{Blank}$$

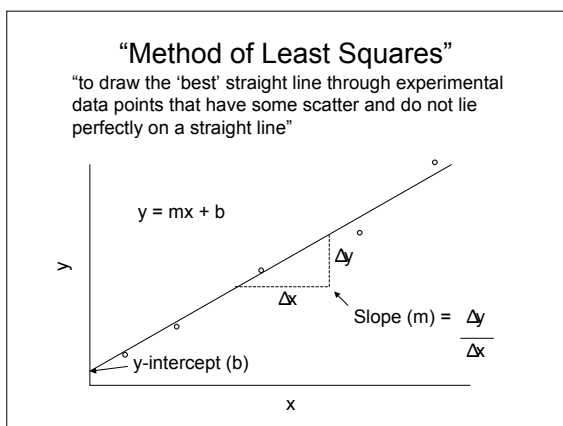
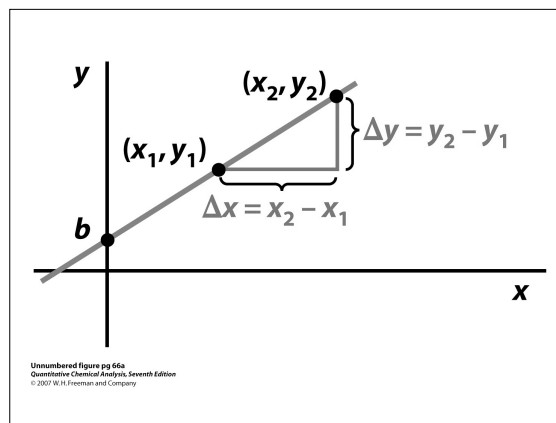
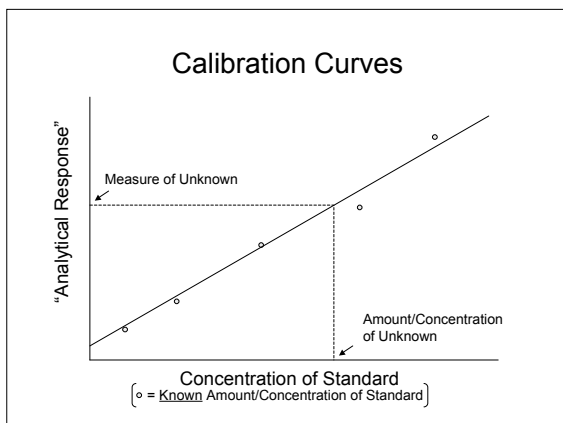
Construction of Calibration Curves

Step 1: Prepare known samples of analyte covering a range of concentrations expected for unknowns. Measure the response of the analytical procedure for these standards.

Step 2: Subtract the average response of the **blank** samples from each measured standard to obtain the corrected value.

Step 3: Make a graph of corrected versus concentration of standard, and use the "method of least squares" procedure to find the best straight line through the linear portion of the data.

Step 4: To determine the concentration of an **unknown**, analyze the unknown sample along with a blank, subtract the blank to obtain the corrected value and use the corrected value to determine the concentration based on your calibration curve.



Method of Least Squares

Vertical Deviation = $d_i = y_i - y = y_i - (mx_i + b)$

$d_i^2 = (y_i - y)^2 = (y_i - mx_i - b)^2$

We wish to minimize the magnitude of the deviations (regardless of sign) so we square the terms. This is where “Method of least Squares” takes its name.

“Method of Least Squares”

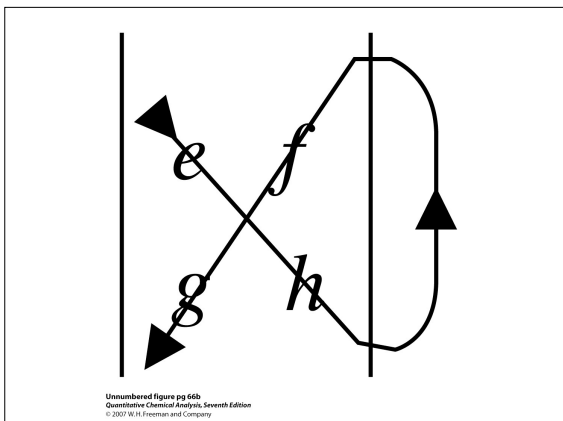
Slope: $m = \frac{\begin{vmatrix} \sum(x_i y_i) & \sum x_i \\ \sum y_i & n \end{vmatrix}}{\begin{vmatrix} \sum x_i^2 & \sum x_i \\ \sum x_i & n \end{vmatrix}} \div D$

Intercept: $b = \frac{\begin{vmatrix} \sum(x_i^2) & \sum(x_i y_i) \\ \sum x_i & \sum y_i \end{vmatrix}}{\begin{vmatrix} \sum x_i^2 & \sum x_i \\ \sum x_i & n \end{vmatrix}} \div D$

$D = \begin{vmatrix} \sum(x_i^2) & \sum x_i \\ \sum x_i & n \end{vmatrix}$

Determinants

$$\begin{vmatrix} A & B \\ C & D \end{vmatrix} \Rightarrow AD - BC$$



“Method of Least Squares”

$$m = \frac{n\sum(x_i y_i) - \sum x_i \sum y_i}{n\sum(x_i^2) - (\sum x_i)^2}$$

$$b = \frac{\sum(x_i^2)\sum y_i - (\sum x_i y_i)\sum x_i}{n\sum(x_i^2) - (\sum x_i)^2}$$

Table 4-6 Calculations for least-squares analysis

x_i	y_i	$x_i y_i$	x_i^2	$d_i (= y_i - mx_i - b)$	d_i^2
1	2	2	1	0.038 46	0.001 479 3
3	3	9	9	-0.192 31	0.036 982
4	4	16	16	0.192 31	0.036 982
6	5	30	36	-0.038 46	0.001 479 3
$\sum x_i = 14$	$\sum y_i = 14$	$\sum(x_i y_i) = 57$	$\sum x_i^2 = 62$		$\sum(d_i^2) = 0.076 923$

Table 4-6
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“Method of Least Squares”

Example: To analyze protein levels, you use a *spectrophotometer* to measure a colored product which results from chemical reaction with protein. To construct a calibration curve, you make the following measurements of absorbance (of the colored product) for several known amounts of protein. Use the “method of least squares” to determine the best fit line.

Amount Protein (mg)	Absorbance	Corrected*
0	0.099	0.000
5.0	0.185	0.086
10.0	0.282	0.183
15.0	0.345	0.246
20.0	0.425	0.326
25.0	0.483	0.384

* Absorbance - Average Blank (=0.099₀)

“Method of Least Squares”

Example: To analyze protein levels, you use a *spectrophotometer* to measure a colored product which results from chemical reaction with protein. To construct a calibration curve, you make the following measurements of absorbance (of the colored product) for several known amounts of protein. Use the “method of least squares” to determine the best fit line.

x_i	y_i	$x_i y_i$	x_i^2
0	0	0	0
5.0	0.086	0.43	25
10.0	0.183	1.83	100
15.0	0.246	3.69	225
20.0	0.326	6.52	400
25.0	0.384	9.60	625
Σ	75	1.225	22.07
Σ	75	1.225	22.07
Σ	75	1.225	22.07

$$m = \frac{n\sum(x_i y_i) - \sum x_i \sum y_i}{n\sum(x_i^2) - (\sum x_i)^2}$$

$$= \frac{(6)(22.07) - (75)(1.225)}{(6)(1375) - (75)^2}$$

$$m = 0.015445714$$

$n = 6 \Rightarrow 6$ data points

“Method of Least Squares”

Example: To analyze protein levels, you use a *spectrophotometer* to measure a colored product which results from chemical reaction with protein. To construct a calibration curve, you make the following measurements of absorbance (of the colored product) for several known amounts of protein. Use the “method of least squares” to determine the best fit line.

x_i	y_i	$x_i y_i$	x_i^2
0	0	0	0
5.0	0.086	0.43	25
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15.0	0.246	3.69	225
20.0	0.326	6.52	400
25.0	0.384	9.60	625
Σ	75	1.225	22.07
Σ	75	1.225	22.07
Σ	75	1.225	22.07

$$b = \frac{\sum(x_i^2)\sum y_i - (\sum x_i y_i)\sum x_i}{n\sum(x_i^2) - (\sum x_i)^2}$$

$$= \frac{(1375)(1.225) - (22.07)(75)}{(6)(1375) - (75)^2}$$

$$b = 0.01109524$$

$n = 6$

"Method of Least Squares"

Example: To analyze protein levels, you use a *spectrophotometer* to measure a colored product which results from chemical reaction with protein. To construct a calibration curve, you make the following measurements of absorbance (of the colored product) for several known amounts of protein. Use the "method of least squares" to determine the best fit line.

$$m = 0.015445714$$

$$b = 0.01109524$$

$$y = (0.015445714)x + (0.01109524)$$

	A	B	C	D	E	F
1	x	y			Formulas:	
2	1	2		slope =		
3	3	3		0.61538	D3 = SLOPE(B2:B5,A2:A5)	
4	4	4		intercept =		
5	6	5		1.34615	D5 = INTERCEPT(B2:B5,A2:A5)	

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	A	B	C	D	E	F	G
1	x	y			Output from LINEST		
2	1	2			Slope	Intercept	
3	3	3		Parameter	0.61538	1.34615	
4	4	4		Std Dev	0.05439	0.21414	
5	6	5		R^2	0.98462	0.19612	Std Dev (y)
6							
7	Highlight cells E3:F5						
8	Type "= LINEST(B2:B5,A2:A5,TRUE,TRUE)"						
9	Press CTRL+SHIFT+ENTER (on PC)						
10	Press COMMAND+RETURN (on Mac)						

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"Method of Least Squares"

"to draw the 'best' straight line through experimental data points that have some scatter and do not lie perfectly on a straight line"

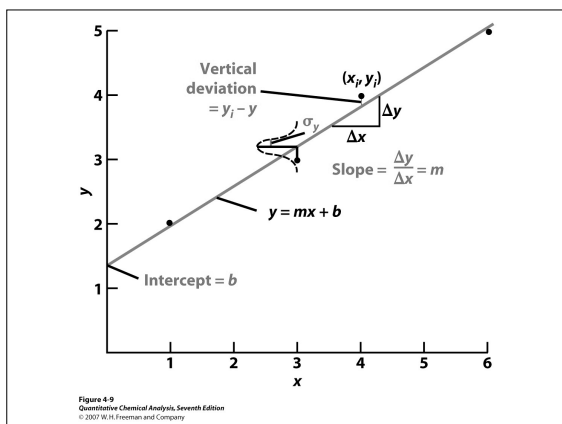
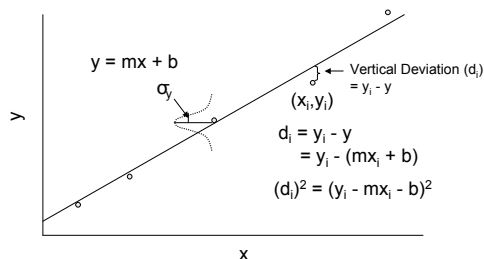


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Uncertainty and Least Squares

$$\sigma_y \approx s_y = \sqrt{\frac{\sum (d_i - \bar{d})^2}{(\text{degrees of freedom})}}$$

$$s_y = \sqrt{\frac{\sum (d_i)^2}{(\text{degrees of freedom})}}$$

$$s_y = \sqrt{\frac{\sum (d_i)^2}{n-2}}$$

Uncertainty and Least Squares

Example: To analyze protein levels, you use a *spectrophotometer* to measure a colored product which results from chemical reaction with protein. To construct a calibration curve, you make the following measurements of absorbance (of the colored product) for several known amounts of protein. Use the "method of least squares" to determine the best fit line. Calculate the uncertainty associated with this line.

x_i	y_i	$x_i y_i$	x_i^2	$d_i (=y_i - mx - b)$	d_i^2
0	0	0	0	-0.011 ₁	0.00012321
5.0	0.086	0.43	25	-0.002 ₂	0.00000540
10.0	0.183	1.83	100	0.017 ₄	0.00030442
15.0	0.246	3.69	225	0.003 ₂	0.00001036
20.0	0.326	6.52	400	0.006 ₀	0.00003589
<u>25.0</u>	<u>0.384</u>	<u>9.60</u>	<u>625</u>	-0.013 ₂	<u>0.00017525</u>
Σ	75.0	1.225	22.07	1375	0.00065442

$n = 6$

Uncertainty and Least Squares

Example: To analyze protein levels, you use a *spectrophotometer* to measure a colored product which results from chemical reaction with protein. To construct a calibration curve, you make the following measurements of absorbance (of the colored product) for several known amounts of protein. Use the "method of least squares" to determine the best fit line. Calculate the uncertainty associated with this line.

$$s_y = \sqrt{\frac{\Sigma(d_i)^2}{n-2}}$$

$$= \sqrt{(0.00065442)/(6-2)}$$

$$= \sqrt{0.0001636}$$

$$= 0.012790808$$

Uncertainty and Least Squares

$$s_m^2 = \frac{s_y^2 n}{D}$$

$$s_b^2 = \frac{s_y^2 \Sigma(x_i^2)}{D}$$

Uncertainty and Least Squares

Example: To analyze protein levels, you use a *spectrophotometer* to measure a colored product which results from chemical reaction with protein. To construct a calibration curve, you make the following measurements of absorbance (of the colored product) for several known amounts of protein. Use the "method of least squares" to determine the best fit line. Calculate the uncertainty associated with this line.

x_i	y_i	$x_i y_i$	x_i^2	d_i^2	
0	0	0	0	0.00012321	
5.0	0.086	0.43	25	0.00000540	
10.0	0.183	1.83	100	0.00030442	
15.0	0.246	3.69	225	0.00001036	
20.0	0.326	6.52	400	0.00003589	
<u>25.0</u>	<u>0.384</u>	<u>9.60</u>	<u>625</u>	<u>0.00017525</u>	
Σ	75.0	1.225	22.07	1375	0.00065442

$n = 6$
 $s_y = 0.012790808$

$$D = \begin{vmatrix} \Sigma(x_i^2) & \Sigma x_i \\ \Sigma x_i & n \end{vmatrix} = \begin{vmatrix} 1375 & 75 \\ 75 & 6 \end{vmatrix} = (1375 \times 6) - (75 \times 75) = 2625$$

Uncertainty and Least Squares

Example: To analyze protein levels, you use a *spectrophotometer* to measure a colored product which results from chemical reaction with protein. To construct a calibration curve, you make the following measurements of absorbance (of the colored product) for several known amounts of protein. Use the "method of least squares" to determine the best fit line. Calculate the uncertainty associated with this line.

x_i	y_i	$x_i y_i$	x_i^2	d_i^2	
0	0	0	0	0.00012321	
5.0	0.086	0.43	25	0.00000540	
10.0	0.183	1.83	100	0.00030442	
15.0	0.246	3.69	225	0.00001036	
20.0	0.326	6.52	400	0.00003589	
<u>25.0</u>	<u>0.384</u>	<u>9.60</u>	<u>625</u>	<u>0.00017525</u>	
Σ	75.0	1.225	22.07	1375	0.00065442

$n = 6$
 $s_y = 0.012790808$, $D = 2625$

$$s_m^2 = \frac{s_y^2 n}{D} = \frac{(0.012790808)^2 (6)}{(2625)} = 0.000000373954$$

$$s_m = 0.000611518$$

Uncertainty and Least Squares

Example: To analyze protein levels, you use a *spectrophotometer* to measure a colored product which results from chemical reaction with protein. To construct a calibration curve, you make the following measurements of absorbance (of the colored product) for several known amounts of protein. Use the "method of least squares" to determine the best fit line. Calculate the uncertainty associated with this line.

x_i	y_i	$x_i y_i$	x_i^2	d_i^2	
0	0	0	0	0.00012321	
5.0	0.086	0.43	25	0.00000540	
10.0	0.183	1.83	100	0.00030442	
15.0	0.246	3.69	225	0.00001036	
20.0	0.326	6.52	400	0.00003589	
<u>25.0</u>	<u>0.384</u>	<u>9.60</u>	<u>625</u>	<u>0.00017525</u>	
Σ	75.0	1.225	22.07	1375	0.00065442

$n = 6$
 $s_y = 0.012790808$, $D = 2625$

$$s_b^2 = \frac{s_y^2 \Sigma(x_i^2)}{D} = \frac{(0.012790808)^2 (1375)}{(2625)} = 0.0000856977$$

$$s_b = 0.009257307$$

Uncertainty and Least Squares

Example: To analyze protein levels, you use a *spectrophotometer* to measure a colored product which results from chemical reaction with protein. To construct a calibration curve, you make the following measurements of absorbance (of the colored product) for several known amounts of protein. Use the "method of least squares" to determine the best fit line. Calculate the uncertainty associated with this line.

$$m = 0.015445714 \pm 0.000611518$$

$$= 0.0154 \pm 0.0006$$

$$b = 0.01109524 \pm 0.009257307$$

$$= 0.011 \pm 0.009$$

Linearity

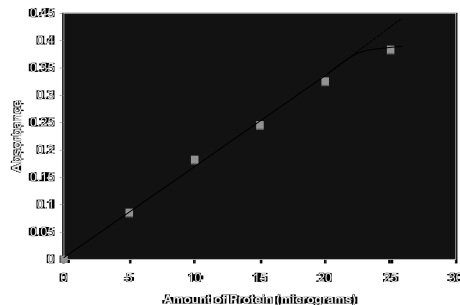


Table 4-7 Spectrophotometer data used to construct calibration curve

Amount of protein (μg)	Absorbance of independent samples			Range	Corrected absorbance		
0	0.099	0.099	0.100	0.001	-0.003 ₃	-0.003 ₃	0.000 ₂
5.0	0.185	0.187	0.188	0.003	0.085 ₇	0.087 ₇	0.088 ₇
10.0	0.282	0.272	0.272	0.010	0.182 ₇	0.172 ₇	0.172 ₇
15.0	0.345	0.347	(0.392)	0.047	0.245 ₇	0.247 ₇	—
20.0	0.425	0.425	0.430	0.005	0.325 ₇	0.325 ₇	0.330 ₇
25.0	0.483	0.488	0.496	0.013	0.383 ₇	0.388 ₇	0.396 ₇

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Linear Range vs. Dynamic Range

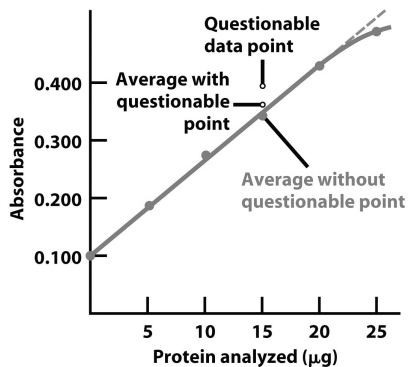
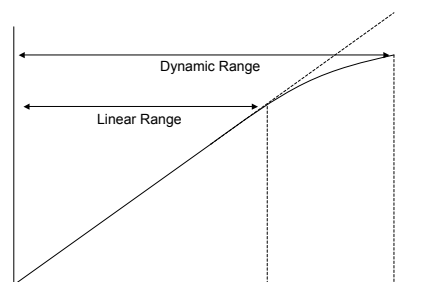


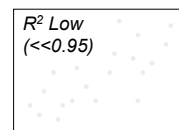
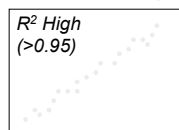
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Determining Linearity

Square of Correlation Coefficient

$$R^2 = \frac{[\sum(x_i - \bar{x})(y_i - \bar{y})]^2}{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}$$

R^2 close to 1
(e.g. $\geq 0.99, 0.98, 0.95$)



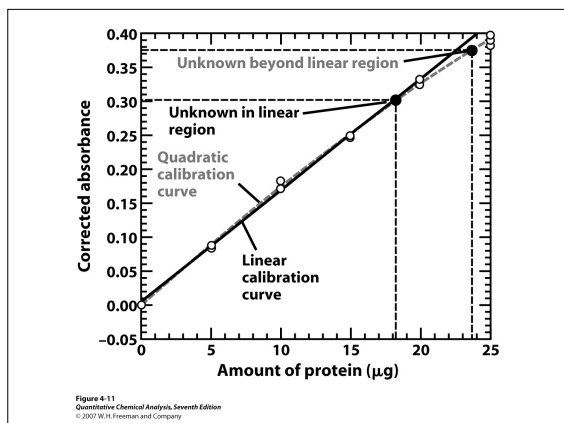


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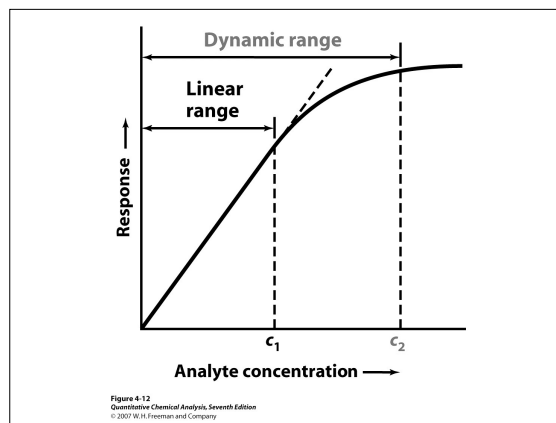


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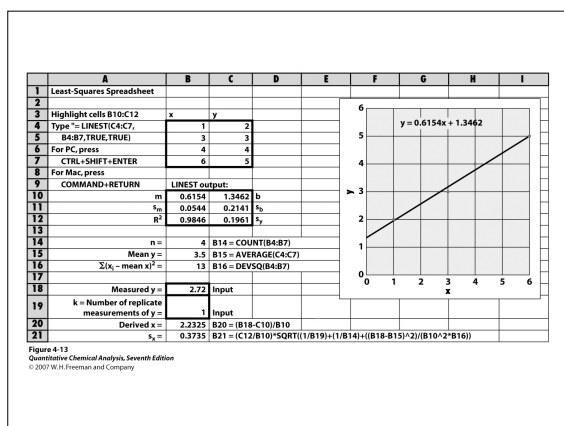


Figure 4-13
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